# Unified Modeling Language (UML) Structural & Behavioral Models

## Academic Grievance, No-Dues & Clearance Automation Engine

**Punjab Engineering College (Deemed to be University), Sector 12, Chandigarh**
**Submitted by:** Raghav Bhardwaj & Karan Jain
**Submitted to:** Dr. Rajesh Bhatia (Course: Software Engineering)

---

### OMG UML 2.5 Modeling Rules & Standards Compliance Matrix

This document conforms strictly to the **Object Management Group (OMG) UML 2.5 Superstructure Specification**:

| Diagram Type | Diagram Classification | Formal OMG UML 2.5 Rules & Semantics Applied | Section |
|---|---|---|---|
| **Use Case Diagram** | Behavioral | Actors outside system boundary; use cases within boundary; precise `<<include>>` (mandatory functional dependency) and `<<extend>>` (conditional extension) stereotypes. | Section 1 |
| **Class Diagram** | Structural | 3-compartment notation; member visibility indicators ( `+` Public, `#` Protected, `-` Private); precise relationship semantics: Generalization (`< | `--`), `Realization (`.. |
| **Activity Diagram** | Behavioral | Concurrency fork and join bars; decision and merge diamonds; action states; initial/final activity nodes; swimlane encapsulation of gate authorities. | Section 3 |
| **Sequence Diagram** | Interaction | Participant lifelines; strictly descending chronological time axis; synchronous calls ( `-->>` ), returns ( `-->>` ); interaction use fragments ( `alt` , `opt` , `loop` ). | Section 4 |
| **State Machine Diagram** | Behavioral | State transitions triggered by events and guards ( `Event [guard] / action` ); composite states ( `Gate 1` , `Gate 2` , `Gate 3` ); initial pseudo-states ( `[*]` ). | Section 5 |
| **Component & Deployment** | Structural / Physical | 3-tier boundary separation (Client UI $\rightarrow$ Application Services $\rightarrow$ Persistent Data Store); interface connectors and protocol definitions. | Section 6 |

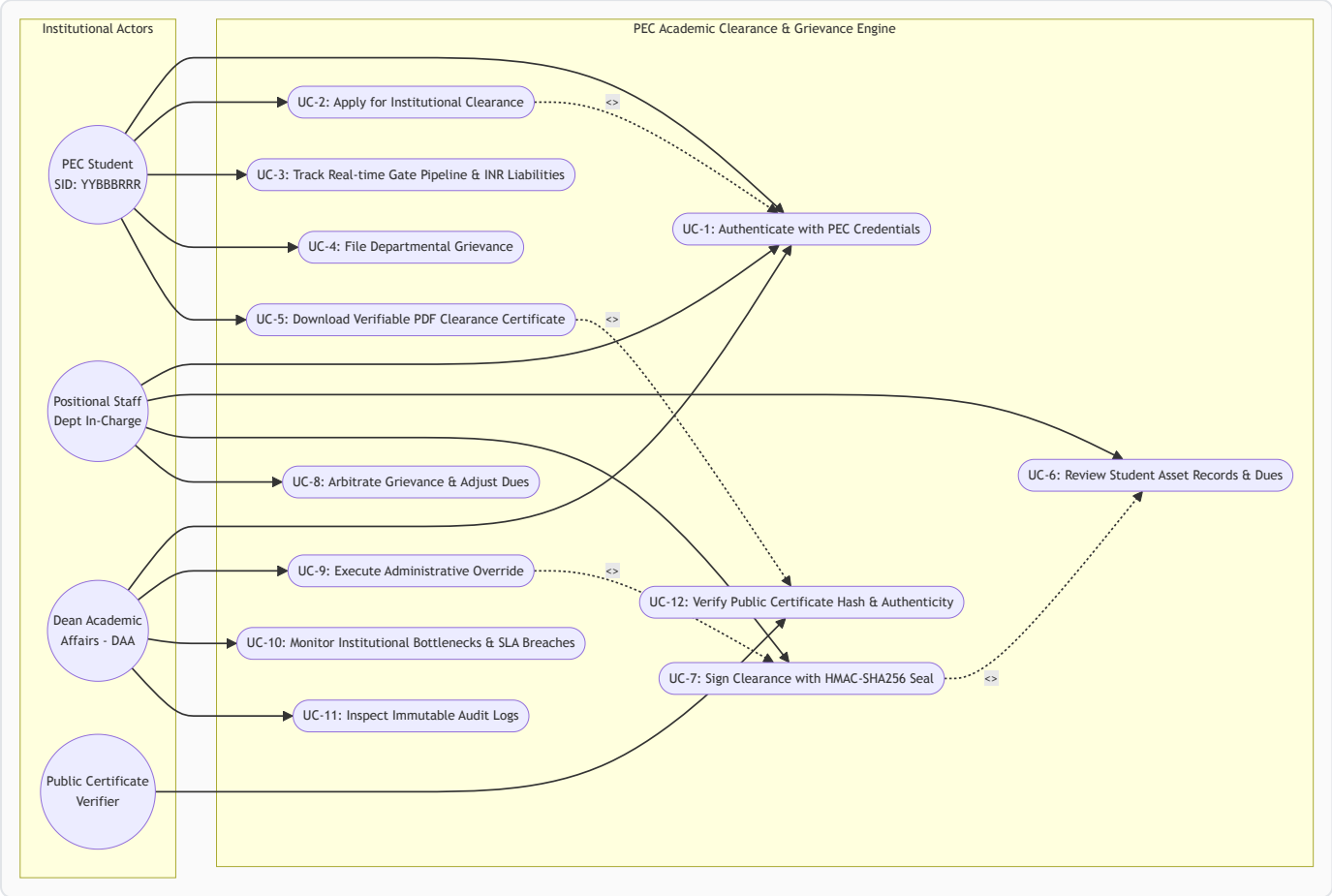
---

# 1. Use Case Model & Diagram

## 1.1 Actors & Role Boundaries

- **Student Actor:** Registered PEC student identified by an 8-digit SID (e.g., `22103042` ).
- **Positional Staff Actor:** Departmental authority operating under permanent institutional designations (*Librarian*, *Chief Warden*, *Gymkhana President*, *HOD Labs*, *Accounts Officer*).
- **Dean Academic Affairs (DAA / Admin Actor):** Executive authority holding override privileges and institutional audit oversight.
- **Public Certificate Verifier Actor:** External unauthenticated party validating clearance credentials.

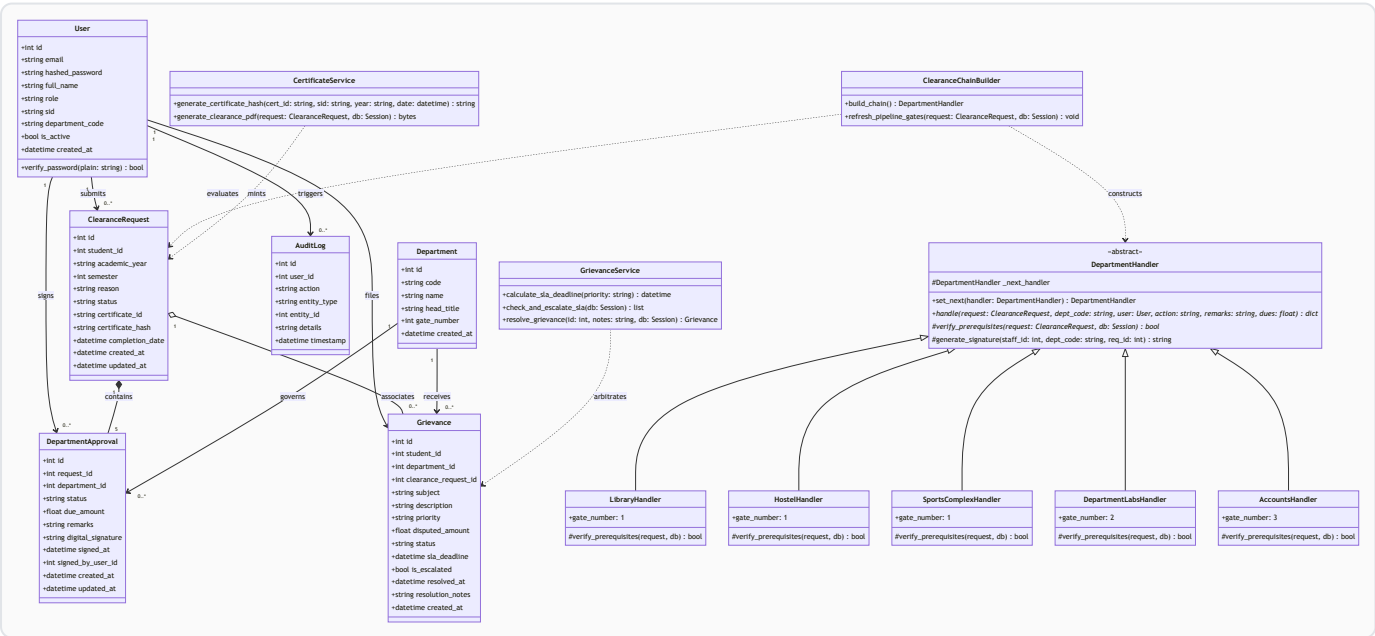
## 1.2 Use Case Diagram




---

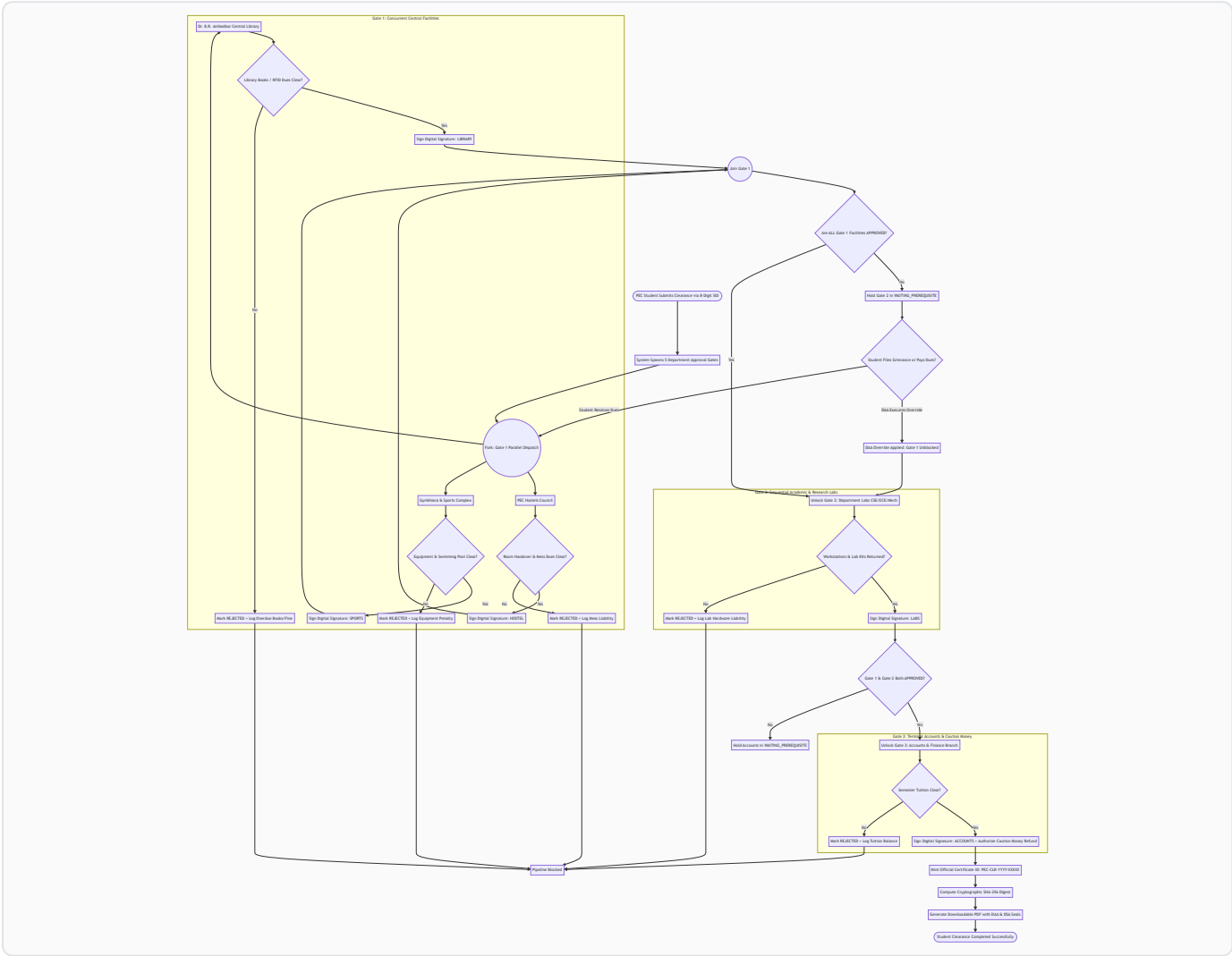
# 2. Static Structural Model: Class Diagram

The class diagram captures the object-oriented domain entities, the Chain of Responsibility behavioral pattern handlers, cryptographic helper utilities, and service orchestration layers:



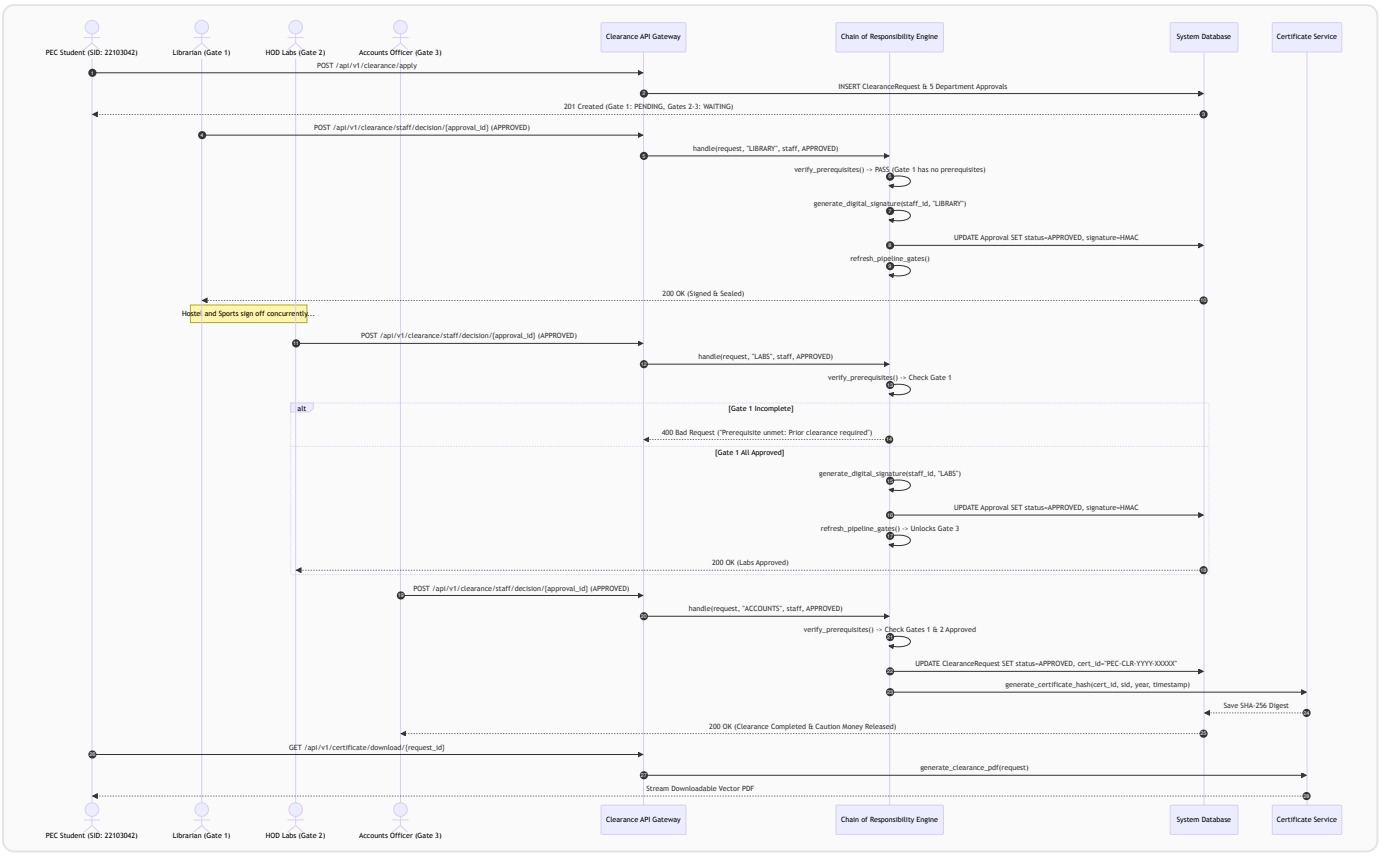
## 3. Dynamic Behavioral Model: Activity Diagram

This activity diagram captures the end-to-end lifecycle of a PEC clearance request, illustrating parallel concurrency forks, barrier joins, and sequential dependency gates:

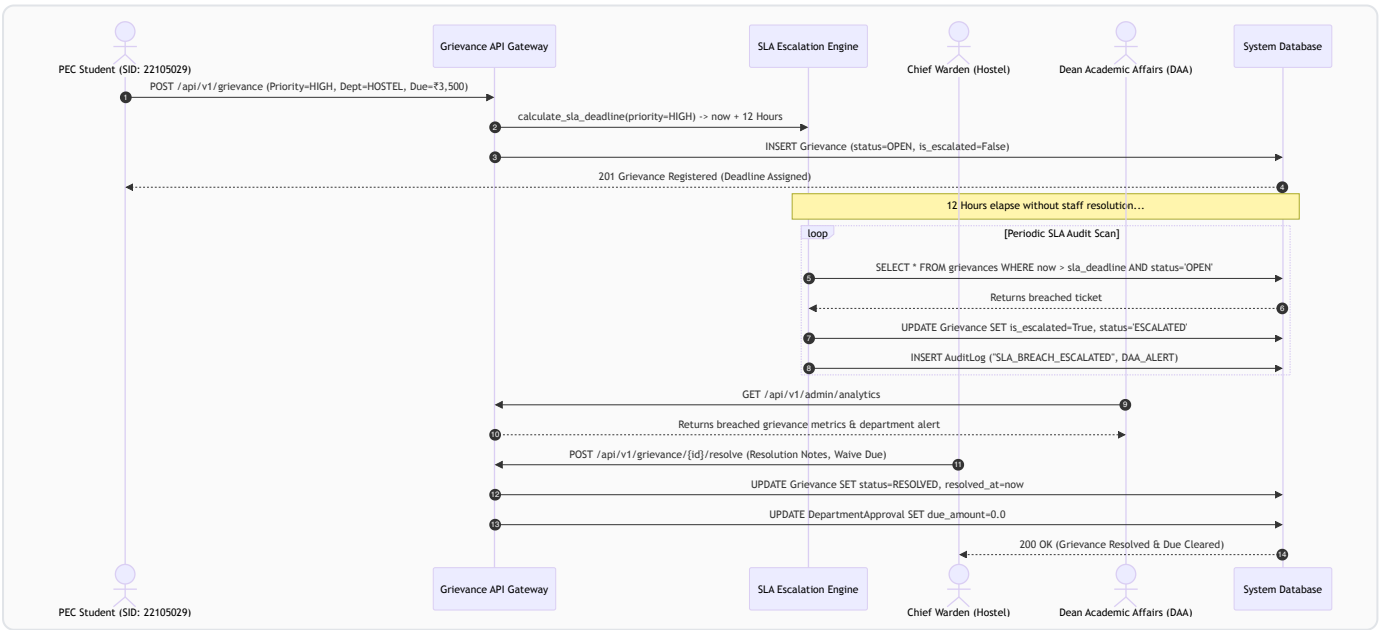


# 4. Interaction Models: Sequence Diagrams

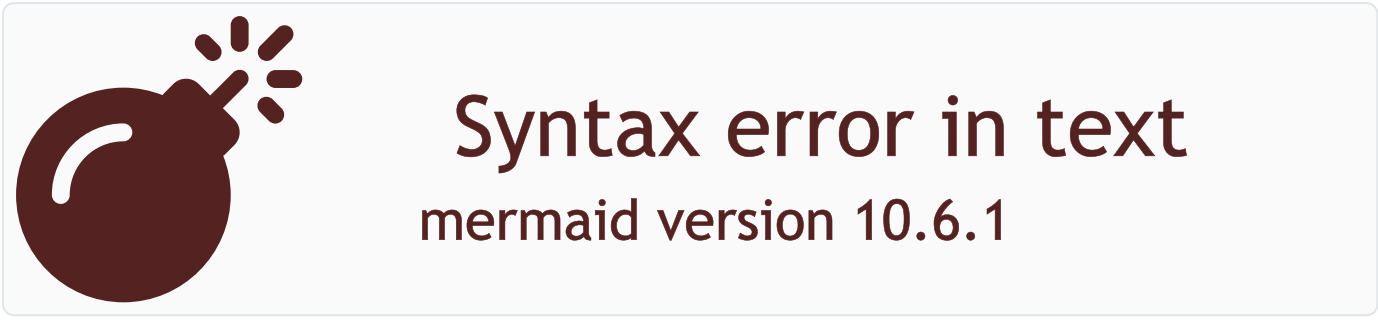
## 4.1 Sequence Diagram 1: Multi-Gate Approval & Cryptographic Minting




## 4.2 Sequence Diagram 2: Grievance Lifecycle & Automated DAA Escalation



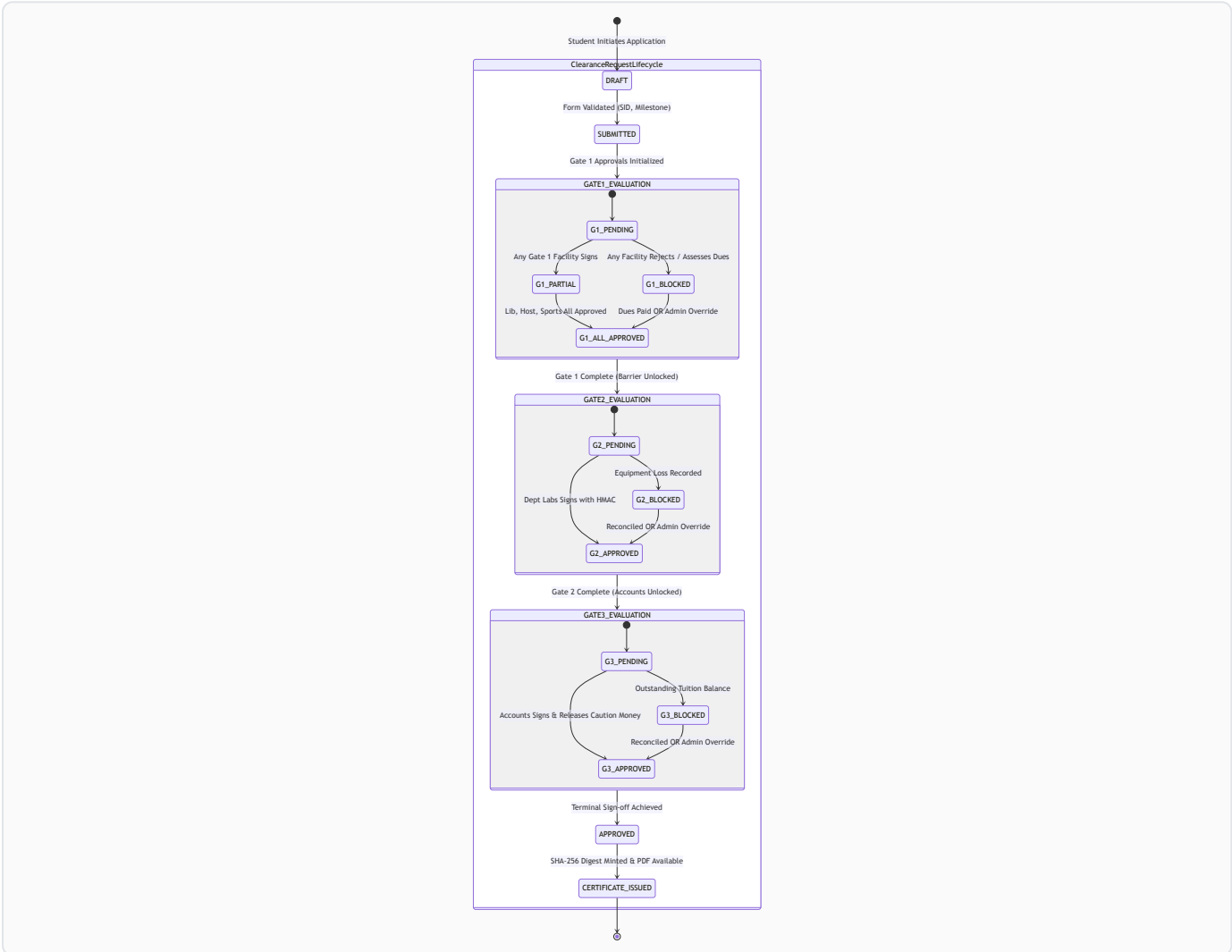
### 4.3 Sequence Diagram 3: Administrative Override & Concurrency Race Prevention




---

## 5. Behavioral State Machine Diagram

This state machine models the discrete status lifecycles of both the master clearance request and individual department approval gates:




---

# 6. System Architecture & Component Deployment Diagram

The component diagram defines the physical and logical boundaries of the system, verifying complete separation between presentation, application services, and persistent storage:

